This is a review submitted to Mathematical Reviews/MathSciNet.

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Title: Gaussian processes with errors in variables: theory and computation.

MR Number: MR4582509

Primary classification:

Secondary classification(s):

Review text:

This article is concerned with the generic regression problem with errors-in-variables in the following form in which responses Y_is are observed corresponding to evaluations of an unknown regression function f on noise-contaminated covariates W_is as

$$Y_i = f(X_i) + \epsilon_i, \quad \epsilon \overset{\text{i.i.d}}{\sim} N(0, \sigma^2)$$
$$W_i = X_i + u_i, \quad , u_i \overset{\text{i.i.d}}{\sim} N(0, \delta^2), \quad X_i \overset{\text{i.i.d}}{\sim} p, \quad i = 1, 2, \cdots, n.$$

with relevant priors specified in the following

$$f \sim \Pi_f, \quad p \sim \Pi_p, \quad \sigma^2 \sim \Pi_{\sigma^2}, \quad \delta^2 \sim \Pi_{\delta^2}.$$

where Y_i is assumed to be conditionally independent of W_i given X_i. In the Bayesian framework, the posterior distribution of unknown parameters $\theta = (f, p, \delta)$ with given observed values $D_n = \{(Y_i, W_i), i = 1, 2, \cdots, n\}$ is obtained by using Bayes rule

$$P(\theta|D_n) = \frac{P(D_n|\theta)P(\theta)}{P(D_n)}.$$

By specifying a Gaussian process prior on the regression function f and a Dirichlet process Gaussian mixture prior on the unknown distribution of the unobserved covariates that is capable of recovering the unknown infinite dimensional parameters f, the authors showed that the posterior distribution of the regression function and the unknown covariate density attain optimal rates of contraction adaptively over a range of Hölder classes, up to logarithmic terms.

In addition, the authors also developed a novel surrogate prior for approximating the Gaussian process prior that leads to efficient computation and preserves the covariance structure, hence, facilitating easy prior elicitation.

Besides the theoretical results, the authors compared several competitors and evaluated their performance based on mean square errors (MSE) and averaged mean squared errors (AMSE) values. The empirical performance of the proposed approach was demonstrated through simulation experiments and a real data example.

Comments to the MR Editors (not part of the Review Text):
The late submission of this review is due to a leave in the past few months.